oxidized oil or varnish were used to waterproof fabrics. Modernfabrics are coated with synthetic polymers.



4.7 Dyeing

Dyeing is the substance which is fixed more or less permanently on the fabric which evokes colour.

4.7.1. Types of Dyes

Dyes are classified according to hue produced chemical class, method of application and the types of fibres to which they are applied. Some of the different dyes include:



Dyes

i. Direct dyes:

Direct dyes colour the fabric directly in a single operation without any affixing agent. Direct dyes are water soluble and are applied mostly to cellulosic fibres. These dyes are dissolved in water and salt is added to control the absorption rate of the dye by the fibre. Then the cloth which is to be dyed is immersed. Direct dyed stuff have relatively excellent light fastness and good colour fastness to sunlight.

Example: Cotton

ii. Acid dyes:

When acidification is done on basic dyes, acid dyes are formed. Acid dyes are used on protein, acrylic and nylon fibres. They have no affinity for cellulosic fibre and are not suitable for fibres which are sensitive to weak acid solutions. They have excellent light fastness and some have good colour fastness to dry cleaning and perspiration.

Examples: Nylon, Silk, Wool

iii. Basic dyes:

Basic or cationic dyes are excellent for colouring acrylic fibres. They are mostly used as "topping" colours to give brilliant colour effects of fabric. Because of the variety of colour effects produced, it is successfully used on modified nylon and modified polyester.

Examples: Wool and Silk

iv. Vat dyes:

Vat dyes have excellent colour fastness property and are suitable on all cellulosic fibres and man-made fibres. It is not suitable on protein fibres because of the alkaline bath which will damage the fibres. There is a wide choice of vat dyes and they withstand hard wear and are fast colours.

Examples: Cotton

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